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Image Compression Model Based on Dynamic Convolution and Vision Mamba

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ABSTRACT

We propose an efficient image compression scheme leveraging Vision Mamba and dynamic convolution, addressing the limitations of existing methods, such as failure to capture long-range pixel dependencies and high computational complexity. Our approach improves both global and local information learning with reduced computational cost. Experimental results on the Kodak, Tecnick and CLIC datasets show that our model achieves competitive performance with lower algorithm complexity. Our code is available on: https://github.com/Lynxsx/ICVM.

1 | Introduction

As a key information carrier, images are crucial for information interaction and transmission. Advances in electronic devices and imaging technology have led to a significant increase in digital image data. To address challenges in image storage and transmission, researchers have developed image compression techniques. Traditional image compression techniques primarily rely on transform coding, which reduces redundancy by transforming images from the spatial domain to the frequency domain and discarding less significant energy while preserving image quality. Classical examples include JPEG [1], based on discrete cosine transform (DCT), and JPEG2000 [2], based on discrete wavelet transform (DWT). Both are widely used image compression methods. Subsequently, newer image compression methods, such as WebP [3], BPG [4] (based on HEVC), VVC [5] (versatile video coding), and AVIF [6] (based on AV1), have emerged. These formats achieve higher compression efficiency and better visual quality than JPEG2000.

In recent years, the development of deep learning technology has led to a surge in interest in learned image compression schemes based on deep learning. Leveraging the powerful learning capabilities of neural networks, researchers can design networks that surpass traditional image compression algorithms under given optimization conditions. Among these, end-to-end image compression schemes show the greatest potential. These frameworks use rate-distortion optimization as a learning criterion, combining bit rate and distortion to reduce the complexity of manual post-processing while making the network more compact and reliable. Notable examples include the recurrent neural network (RNN)-based framework by Toderici et al. [7] and the convolutional neural network (CNN)-based framework by Ballé et al. [8].

Toderici et al. [7] pioneered the use of RNNs for end-to-end image compression, incorporating long short-term memory (LSTM) [9] and gated recurrent units (GRUs) [10] to enable context-based entropy coding, achieving superior compression performance. Inspired by variational autoencoders (VAEs) [11], Ballé et al. [8] employed CNNs for multiple down-sampling, mapping image data into a latent space for encoding and decoding. Additionally, they introduced a generalized divisive normalization (GDN) [12] layer to replace traditional batch normalization (BN) [13]. To address the non-differentiability of scalar quantizers, Ballé et al. [8] proposed adding uniform noise instead of quantization, enabling backpropagation during training.

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However, these image compression schemes face several limitations. For instance, they treat all pixels equally, failing to prioritize important information as the human visual system does, which hinders feature learning performance. To address this, Cai et al. [14] introduced a region-of-interest (ROI) mechanism into the VAE-based framework, enabling higher-quality compression in key regions by allocating more computational resources. Ma et al. [15] combined deep learning with wavelet transform, designing a wavelet-like transform method for compression. Meanwhile, Li et al. [16] and Tang et al. [17] incorporated attention mechanisms to enhance feature extraction and achieve better results. Inspired by the superior performance of Transformer [18] over convolutions, researchers have integrated Transformer-based architectures [19–21] into compression networks. Others focused on entropy coding, developing advanced entropy models to boost network expressiveness [22, 23]. Additionally, methods like fractal image compression [24] and adaptive techniques [25] reduce redundancy through preprocessing, further improving compression performance. Recently, an adversarial steganography detection scheme [26] was proposed, leveraging preprocessing and feature migration technologies. By uniformly sampling patch images from regions with varying steganographic modification probabilities, the approach effectively addresses the sparsity of steganographic and adversarial embeddings.

Although existing methods improve network compression performance, they face significant challenges, particularly in computational cost. Smaller networks reduce resource consumption but often underperform, while deeper networks increase hardware burdens during training and inference, limiting their practicality. Therefore, some researchers have explored implicit neural representations (INRs) [27–29], which map pixel positions to RGB values. INRs avoid quantization, enabling direct optimization of rate-distortion performance. This approach opens up new possibilities for image compression and demonstrates significant potential. However, these methods require training on the target image to achieve optimal performance, which limits their generalization capability.

To overcome the above challenges on the VAE-based framework, we designed a residual neural network based on Vision Mamba [30] and omni-dimensional dynamic convolution (OD Conv) [31]. This network leverages Vision Mamba to enhance global modelling capabilities without significantly increasing computational complexity. Additionally, it incorporates OD Conv and residual skip-connections [32] to improve the network's ability to focus on crucial information. Our contributions can be summarized as follows:

- We propose a Vision Mamba layer tailored for image compression, which captures long-range pixel dependencies through a four-directional scan. Based on our investigation, our work is one of the early attempts to adapt Vision Mamba for image compression, demonstrating its potential in this field.
- We integrate OD Conv and residual skip-connections to construct a down/up-sampling layer. The designed residual convolutional layer is plug-and-play, improving the network's feature extraction capability.
- Building on the above innovations, we develop a learned image compression framework based on VAE, designed as a lightweight model. Additionally, we introduce an improved channel-wise attention mechanism into the entropy model, enabling the construction of a heavyweight model for higher performance.
- According to extensive experimental results, our proposed lightweight model achieves excellent performance comparable to existing models with lower algorithm complexity; the heavyweight model, at the cost of increased complexity, achieves even better performance.

The rest of the paper is organized as follows. Section 2 reviews related work. Section 3 describes our proposed model in detail. Experimental results and analyses are presented in Section 4, and the work is summarised in Section 5.

2 | Related Work

2.1 | Convolutional and Residual Network

In most learned image compression schemes, convolution [33] plays a crucial role. By sliding convolutional kernels over images, neural networks capture structural, textural, and detailed features for fusion. Early end-to-end image compression frameworks relied on a limited number of regular convolutions for down-sampling and dimension boosting [8], often resulting in information loss during the mapping of image data into the latent space, which negatively impacted compression performance. To address this, researchers have proposed various enhancements to convolutional layers, such as using residual blocks [34] or dense blocks [35] as the network backbone. Others have combined asymmetric convolutions [17] with regular convolutions, enabling 1×3, 3×1, and 3×3 convolutions to operate simultaneously, enriching feature extraction in both vertical and horizontal directions.

However, these methods often increase network depth, leading to higher computational complexity and making deployment on average-performance hardware platforms challenging. Through extensive experiments, we found that residual networks improve performance, but stacking too many convolutional layers or adding excessive residual skip-connections significantly increases complexity. Recent studies [36] suggest that plug-and-play convolutions offer both flexibility and control over network complexity. Inspired by this, we designed a plug-and-play residual convolutional layer that achieves a better balance between computational complexity and performance.

2.2 | Vision Mamba Model

Transformer [18], a deep learning model based on the self-attention mechanism, excels at capturing long-range dependencies between any positions in input sequences. It has demonstrated superior performance compared to RNNs and CNNs in natural language processing (NLP). However, Transformer suffers from high computational complexity, requiring more time for training and inference, as well as significant hardware resources. Recently, Albert et al. [37] proposed a sequence model called

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'Mamba', which builds on state space models. By incorporating a selection mechanism and a hardware-aware algorithm, Mamba updates hidden states based on input sequences and predicts output sequences. While achieving performance comparable to Transformer, Mamba reduces computational complexity from quadratic $O\left(N^2\right)$ to linear $O(N)$. Vision Mamba [30], an extension of Mamba for 2D image processing, has achieved competitive results with Vision Transformer [38] in computer vision (CV) tasks [30, 39]. To better model image data by leveraging contextual content and the interaction of spatial and channel information, we designed a Vision Mamba module specifically adapted for image compression. This module achieves competitive performance while maintaining hardware-friendly characteristics by effectively controlling computational complexity.

2.3 | VAE-Based Framework

A VAE-based network [11] typically consists of an encoder–decoder pair. The encoder maps the input raw data x to a latent representation x', which captures the essential features of the raw data. The decoder then reconstructs x' into $\hat{x}$. We illustrate three mainstream compression frameworks in Figure 1. Figure 1a shows the first CNN-based image compression model proposed by Ballé et al. [8]. Figure 1b depicts the improved model by Ballé et al. [40], which introduces a hyperprior module to model pixel distributions using Gaussian assumptions for high-quality entropy coding. Further enhancements by Minnen et al. [41] include a contextual model and non-zero Gaussian distribution modelling, as shown in Figure 1c).

As shown in Figure 1a, the CNN-based end-to-end image compression framework proposed by Ballé [8] consists of a nonlinear analysis transform g_a (labelled as 'analysis transform') and a nonlinear synthesis transform g_s (labelled as 'synthesis transform'). Here, X denotes the input raw image, and $\hat{X}$ represents the reconstructed image after compression. The raw data X is mapped into the latent space through g_a, producing the latent representation Y. The quantizer Q then processes Y to obtain $Q(Y)$, which is further compressed through entropy coding. This process can be expressed as:

$$Q(Y) = Q(g_a(X;\theta_a)), \tag{1}$$

where θ_a represents the learnable parameters in the nonlinear analysis transform. Similarly, the decompressed image is reconstructed by decoding the bitstream and passing through g_s, with θ_s denoting the learnable parameters in the nonlinear synthesis transform. This process can be represented as:

$$\hat{X} = g_s\left(\hat{Y};\theta_s\right) = g_s(Q(g_a(X;\theta_s));\theta_a)\,, \tag{2}$$

where $\hat{Y}$ is the result of Y after quantization by the quantizer.

A major advantage of the end-to-end image compression framework is its ability to jointly optimize rate-distortion trade-offs, allowing the neural network to learn the better result by minimizing the loss function:

$$L = R + \lambda D, \tag{3}$$

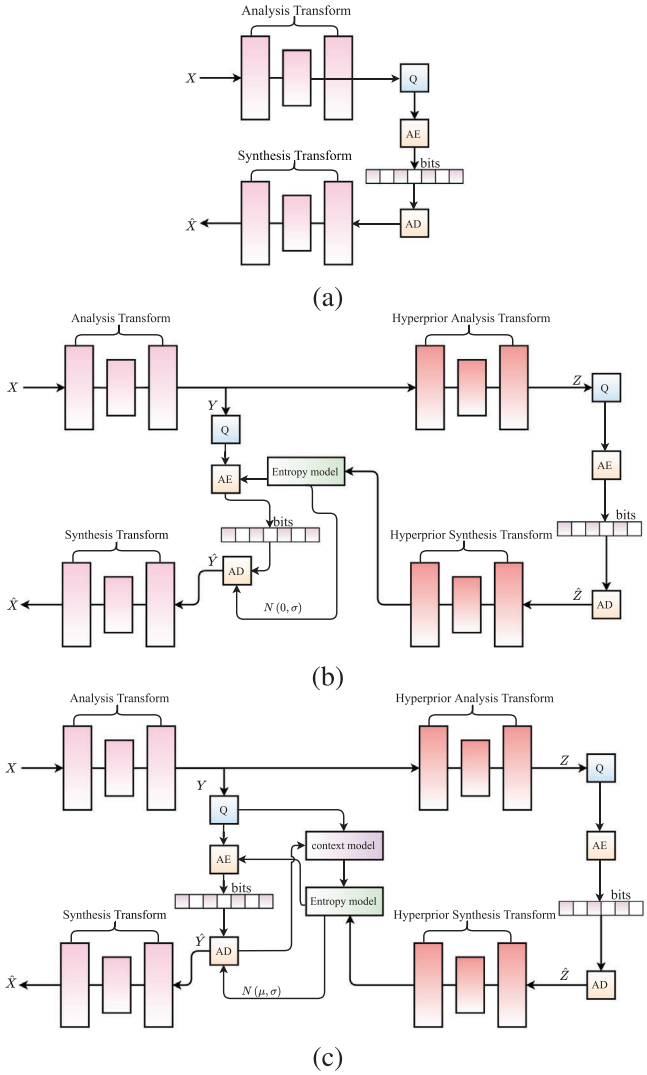


FIGURE 1 | Basic VAE-based image compression framework. (a) Ballé et al. [8] proposed the original framework. (b) Original model with hyperprior [40]. (c) Minnen et al. [41] proposed framework with hyperprior and contextual model.

where R is the bit rate, representing the size of the transmitted data and measuring the degree of compression; D is the distortion, measuring the difference between the reconstructed image $\hat{X}$ and the original image X; and λ is a hyperparameter, analogous to the quality factor in JPEG compression, used to control the compression rate. D is computed as:

$$D = d\left(X,\hat{X}\right), \tag{4}$$

where the function d is typically an objective metric, such as mean squared error (MSE), or perceptual metric like multi-scale structural similarity (MS-SSIM) [42].

The bit rate R is measured by the information entropy of the latent distribution Y, defined as:

$$R = H(p(Y),q(Y)) = E_{y\sim p(y)}(-\log q(Y)), \tag{5}$$

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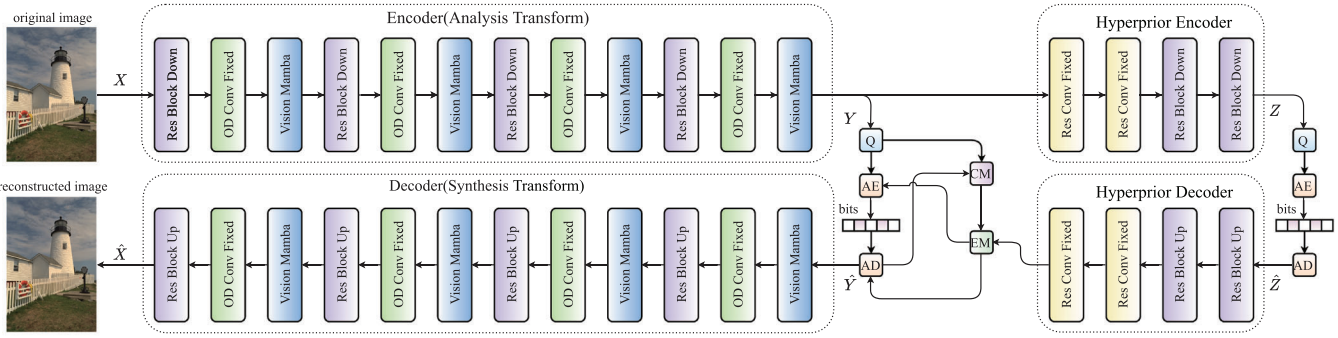


FIGURE 2 | Overview of our model. Q is the quantizer. The network consists mainly of a pair of encoder and decoder, a pair of hyperprior encoder and hyperprior decoder. AE and AD are entropy encoder and entropy decoder. EM and CM denote entropy model and contextual model.

where $p(Y)$ is the true probability distribution of Y, and $q(Y)$ is a neural network-approximated distribution. Since $p(Y)$ is difficult to compute, $q(Y)$ is used as a substitute. In the original model [8], $p(Y)$ was unspecified. To address this, Ballé et al. [40] introduced a hyperprior framework, as shown in Figure 1b, which includes a hyperprior encoder h_a and hyperprior decoder h_s (labelled as 'hyperprior analysis transform' and 'hyperprior synthesis transform'). This framework models each element of Y as a separate zero-mean Gaussian distribution, significantly reducing redundancy in the latent representation. Minnen et al. [41] further extended this approach by incorporating non-zero-mean Gaussian modelling and a contextual model to capture input data correlations, as illustrated in Figure 1c. These enhancements improve the model's expressive power by guiding the generation of mean and variance parameters.

Based on existing research, the VAE-based end-to-end image compression model [8, 40, 41] demonstrates stable performance. To address the long-range dependency limitations of current models, we introduce Vision Mamba, which enhances the encoder's and decoder's learning capabilities while maintaining manageable complexity. Additionally, we combine residual networks and dynamic convolution as the backbone to improve the model's ability to capture key features.

3 | Proposed Network

3.1 | Network Architecture

Our proposed image compression network, based on Vision Mamba and dynamic convolution, is implemented within the VAE compression framework. The backbone consists of up/down-sampling layers, dynamic convolution layers, and Vision Mamba layers, stacked sequentially four times, as shown in Figure 2. The Vision Mamba layer captures long-range dependencies in the input data, enhancing the network's global feature modelling capabilities while maintaining computational efficiency. The dynamic convolution layer employs dynamically weighted kernels, adapting the convolutional kernel weights based on input data features to better align with the network's learning objectives. The down-sampling layer scales the latent representation, mapping raw data into a low-dimensional space for compression, while the up-sampling layer performs the inverse operation.

Our network framework comprises three main modules:

- Res Block Down/Up: This down/up-sampling layer, labelled as 'Res block down/up' in Figure 2, integrates residual skip-connections and channel attention mechanisms to achieve effective feature extraction while avoiding significant increases in computational complexity.
- OD Conv Fixed: An omni-dimensional dynamic convolution layer, labelled as 'OD Conv Fixed' in Figure 2, is embedded throughout the network backbone to further refine extracted features.
- Vision Mamba Layer: The 'Vision Mamba' layer, shown in Figure 2, captures long-range pixel dependencies, enhancing the network's global modelling capabilities.

Notably, the 'Res Conv Fixed' component in the hyperprior encoder/decoder (Figure 2) shares a similar structure with 'Res Block Down/Up' but does not perform down/up-sampling. All modules are designed to be plug-and-play, enabling flexible network adjustments and facilitating future improvements. This design leaves ample room for further exploration. In the following sections, we will introduce these three well-designed layers in detail.

3.2 | Residual Up/Down-Sampling Layer

Figure 3 illustrates the structure of the proposed up/down-sampling layers. In learned image compression, convolution with a stride of 2 is commonly used for down-sampling and dimension expansion. However, relying solely on a single layer of regular convolution limits the receptive field and makes it difficult to capture global information. To address this, we design a down-sampling layer based on residual skip-connections, as shown in Figure 3a. The down-sampling operation uses a regular 3×3 convolution with a stride of 2, which halves the spatial dimensions of the input features while expanding the channels to N dimensions (or M channels for the fourth down-sampling). We employ LeakyReLU [43] as the activation function to accelerate model convergence. Following the down-sampling, a regular 3×3 convolution with a stride of 1 (Conv fixed) is applied to further fuse local features and enhance the model's spatial local modelling capability.

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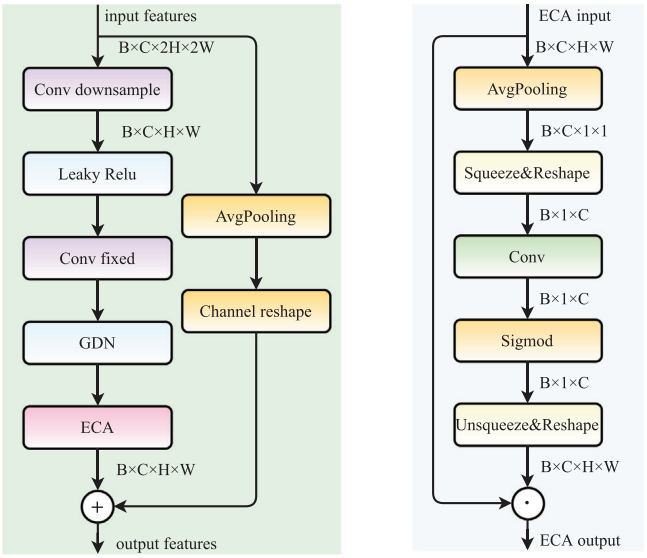


FIGURE 3 | (a) Illustration of down-sampling layer. (b) Illustration of ECA.

Consistent with mainstream image compression frameworks, we replace the batch normalization (BN) layer [13] with the generalized divisive normalization (GDN) layer [12] for normalization. Since the input features are expanded to higher dimensions, we introduce a lightweight channel attention mechanism (ECA) [44] at the end of the down-sampling layer. This mechanism learns the relationships between different channels to determine the contribution of each channel to the final output. The structure of the ECA module is shown in Figure 3b. After applying channel attention, the weighted features are combined with the original inputs via residual skip-connections, ensuring that the network accurately captures the most relevant information. By integrating regular convolution, residual skip-connections, and channel attention, this down-sampling module effectively focuses on important spatial and channel features, improving performance without significantly increasing complexity.

3.3 | OD Convolutional Layer

Regular convolution [33] uses fixed weights during feature extraction, which limits its ability to adapt to varying input features. While smaller convolution kernels reduce computational complexity, they may struggle to capture sufficient contextual information due to their limited receptive field. Conversely, larger convolution kernels can capture more contextual information but at the cost of significantly increased computational complexity. Dynamic convolution addresses this limitation by adaptively adjusting the weights of the convolution kernel based on the input features, enabling the model to focus on the important information in the input.

We designed a plug-and-play dynamic convolution layer based on the omni-dimensional dynamic convolution (OD Conv) proposed by Intel [31]. The structure of the layer is shown in Figure 4a. OD Conv is a full-dimensional dynamic convolution block that employs three types of attention mechanisms to dynamically adjust the convolution weights, enhancing its feature extraction capability while maintaining computational efficiency. OD Conv lies in its ability to adaptively adjust the convolution weights based on the input features through spatial attention and kernel attention. These attention mechanisms work in a complementary manner to capture rich contextual information. In Figure 4a, we use two ODConv blocks to process the input features and add residual skip-connections to combine the inputs and outputs, making the network easier to train and optimize.

Figure 4b demonstrates OD Conv's processing of input features. It assigns different types of attention weights to the convolutional kernel, namely spatial dimension α_{si}, input channel dimension α_{c1}, output channel dimension α_{c2}, and kernel dimension α_{wi}. These attention weights are dynamically generated based on the input features, enabling the convolution operation to adaptively focus on the most important information:

- Spatial dimension α_{si}: It adjusts the importance of each spatial location in the convolution kernel, as illustrated in Figure 4c.
- Channel dimension α_{c1}, α_{c2}: It adjusts the importance of each input/output channel.
- Kernel dimension α_{wi}: It adjusts the importance of multiple convolution kernels, as illustrated in Figure 4d.

The computation of OD Conv can be divided into two main steps, as shown in the following equations:

$$W_d = \sum_{i=1}^{n} \alpha_{wi} \odot \alpha_{si} \odot W_i, \tag{6}$$

$$y = \alpha_{c2} \odot (W_d(\alpha_{c1} \odot x)), \tag{7}$$

where $\alpha_{wi} \in \mathbb{R}^{k\times k}$, $\alpha_{si} \in \mathbb{R}^{k\times k}$, $\alpha_{c1} \in \mathbb{R}^{c_{in}}$ and $\alpha_{c2} \in \mathbb{R}^{c_{out}}$, k represents the size of each dynamic convolution kernel.

OD Conv first computes the spatial attention α_{si} and kernel attention α_{wi} based on the input features. These attention weights are used to dynamically adjust the parameters of each convolution kernel W_i, generating the adjusted convolution kernel W_d. This process allows the convolution operation to adaptively focus on important spatial locations and kernel combinations.

Before applying the adjusted convolution kernel W_d, the input features x are weighted by the input channel attention α_{c1}. After the convolution operation, the output features are further weighted by the output channel attention α_{c2}. This two-stage channel weighting enhances the feature representation by emphasizing important input and output channels.

Such a multi-dimensional dynamic convolution layer enhances the feature extraction capability of the network without introducing a large number of additional parameters. In our designed OD Conv layer, the input and output channels are kept the same, which reduces accuracy degradation due to dimension changes and avoids unnecessary computational overhead.

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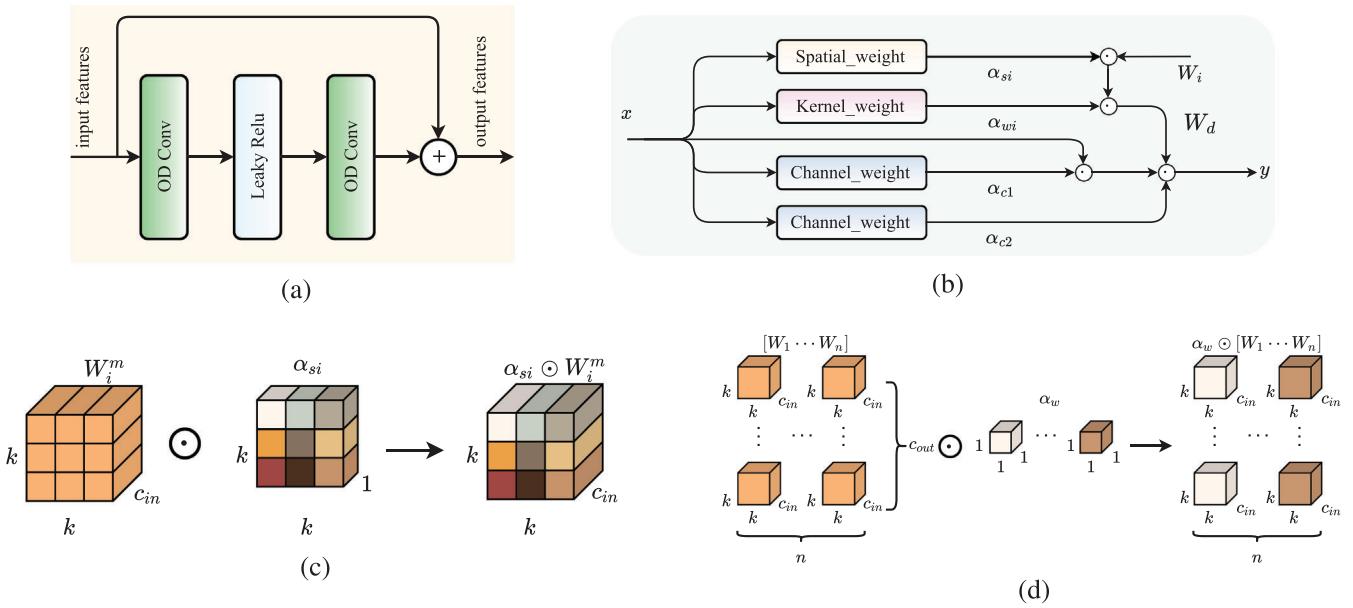


FIGURE 4 | (a) ODConv layer architecture. (b) ODConv operation mechanism. (c) Spatial attention: Adjusting the importance of each spatial location in the convolution kernel. (d) Kernel attention: Adjusting the importance of multiple convolution kernels.

3.4 | Vision Mamba Layer

To enhance the network's ability to model global information while controlling computational complexity, we introduce the Vision Mamba layer. To the best of our knowledge, we are one of the earliest attempts to apply Mamba [37] in the field of image compression. Mamba [37] is derived from structured state-space sequence models [45]. It incorporates selection mechanism and hardware-aware algorithm, enabling it to achieve a competitive effect with Transformer in the processing of long-range one-dimensional data such as speech and text. Liu et al. [30] extended the Mamba model to the vision domain by introducing a simple yet effective approach: They applied a selective scanning mechanism to read image data sequences in a four-way scan and modelled them using a structured state-space sequence model. This adaptation allows Vision Mamba to compete with popular current vision models such as ResNet [32], Vision Transformer [38], Swin Transformer [46], and ConvNet [47] at low computational complexity, demonstrating its strong potential of Vision Mamba in the vision tasks.

The core of Vision Mamba is state space model (SSM) [48], as shown in Figure 5a. SSM can be viewed as a linear time-invariant system that maps an input sequence to an output sequence through hidden states. This process is represented by a system of linear ordinary differential equations (ODEs):

$$h'(t) = Ah(t) + Bx(t), \tag{8}$$

$$y(t) = Ch(t) + Dx(t), \tag{9}$$

where $x(t)$ is the input sequence, and $h(t)$ is the hidden state. $A \in \mathbb{C}^{N\times N}$, $B \in \mathbb{C}^N$ and $C \in \mathbb{C}^N$ are weighting parameters. Since (8) and (9) describe a continuous system, a zero-order hold (ZOH) discretization method computation process is applied to adapt the equations for processing discretized image information. The discretization parameters are obtained as:

$$h(t + \Delta) = \bar{A}h(t) + \bar{B}x(t), \tag{10}$$

$$\bar{A} = \exp(\Delta A), \tag{11}$$

$$\bar{B} = (\Delta A)^{-1}\left(e^{\Delta A} - I\right)\Delta B. \tag{12}$$

In (10), Δ is the time scale parameter, which can be interpreted as the step size. $\bar{A}$ and $\bar{B}$ are the discretized weight parameter matrices. This iterative update process for the state inputs and outputs can be analogized to a recurrent convolutional network, as illustrated in Figure 5a. To convert 2D image data into a 1D sequence suitable for the Mamba model while preserving image context, Vision Mamba unfolds the image data into sequences by means of a four-way selective scan. This approach allows each pixel in the image efficiently integrate information from all directions, as shown in Figure 5b.

Figure 6a shows the proposed Vision Mamba layer. It offers high flexibility, as the number of VSSM blocks can be adjusted to trade off between complexity and performance. Each VSSM block performs Vision SSM mapping computation on the input sequences and employs the channel attention mechanism to capture the relationship between output feature dimensions.

For the input features, $F_{\text{in}} \in \mathbb{R}^{B\times C\times H\times W}$ in each VSSM block will flatten out four sequences $x_{t_i} \in \mathbb{R}^{B\times L\times C}$, $i = 1, 2, 3, 4$ in each of the four directions. Each sequence is sequentially processed by the Vision SSM according to (9) and (10). Additionally, a scaling factor scale $\in \mathbb{R}^C$ is introduced to adjust the inter-channel relationship. The computational flow of the VSSM block is defined as

$$F_v = \text{Vision SSM}(\text{LN}(F_{\text{in}})) + \text{scale} \cdot F_{\text{in}}, \tag{13}$$

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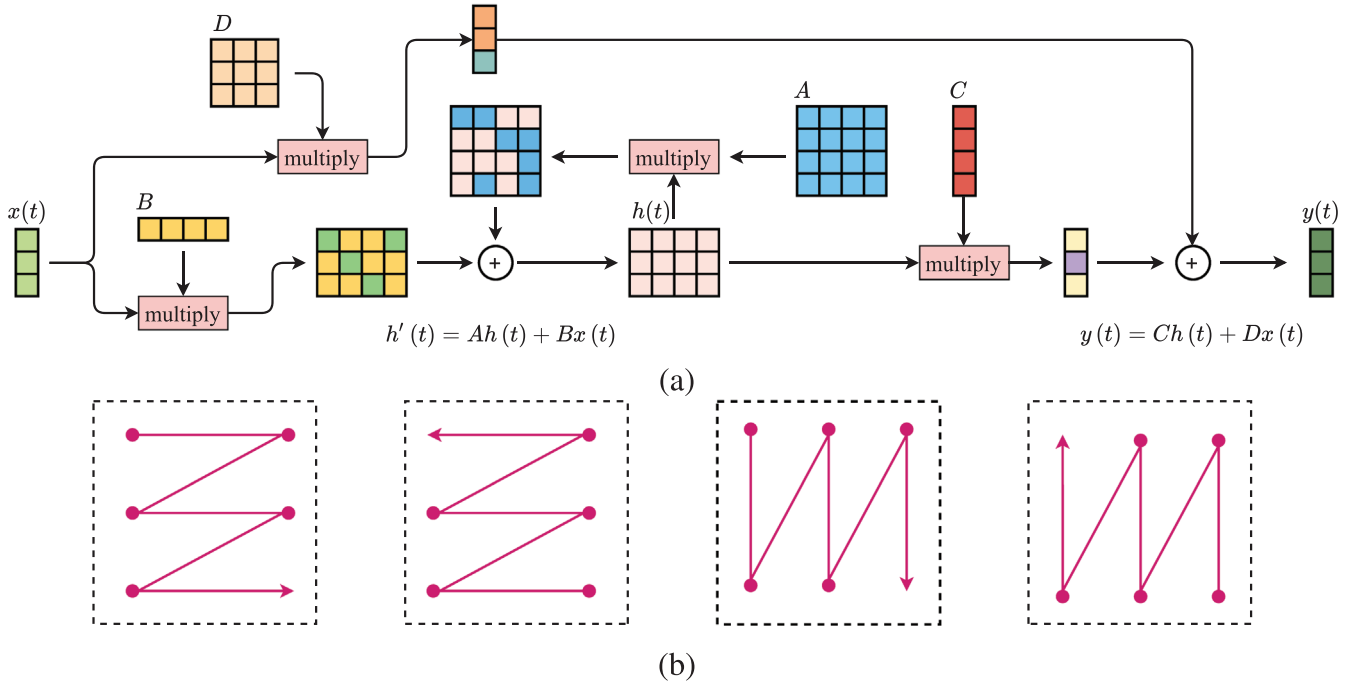


FIGURE 5 | (a) An iterative process of mapping inputs to outputs and updating hidden states in Mamba models. (b) Four directions of spreading image data for selective scanning mechanisms.

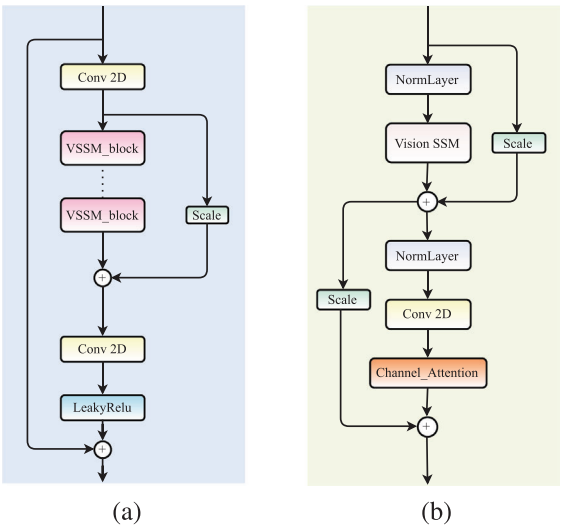


FIGURE 6 | (a) Illustration of our proposed Vision Mamba layer. (b) Illustration of VSSM block.

$$F_{\text{out}} = \text{CA}(\ \text{LN}(\text{Conv}(F_v))) + \text{scale} \cdot F_v, \tag{14}$$

where $F_v \in \mathbb{R}^{B\times C\times H\times W}$, $F_{\text{out}} \in \mathbb{R}^{B\times C\times H\times W}$ are the output of the features, Vision SSM refers to the "Vision SSM" module in Figure 6b, LN denotes the layer normalization, and CA represents the channel attention mechanism. VSSM block preserves the dimensions of the inputs and outputs, similar to the attention mechanism.

Our contribution lies in the redesign of the Vision Mamba layer to better suit image compression tasks. Unlike implementations that rely on complex stacking and nesting of modules, we simplify the architecture by retaining only the essential VSSM block for long-range dependency modelling. The rest of the module is constructed using minimal components, including a few convolutional layers, residual skip-connections, and a channel attention mechanism. This design ensures that the module focuses on learning long-range dependencies while maintaining consistent input and output tensor dimensions throughout the processing.

By reducing redundancy and focusing on the core functionality of long-range dependency modelling, our Vision Mamba layer achieves a better balance between performance and computational efficiency. This makes it particularly effective for image compression, where preserving global structures and fine details is critical. Extensive experiments demonstrate that it can improve the performance of image compression without significantly increasing the computational complexity.

3.5 | Channel-Wise Entropy Model

Recent research [49] has shown that improvements to entropy models can significantly enhance image compression performance. Conventional entropy models use 1×1 convolutions to predict the mean and variance in the latent space. While this method is simple and has minimal impact on model complexity, it overlooks inter-channel relationships and fails to accurately capture cross-channel correlations, leading to less precise probability distribution predictions. Recently, Li et al. [50] proposed a channel-wise entropy model inspired by the Vision Transformer [38], which improves prediction accuracy at the cost of increased complexity. Its framework is illustrated in Figure 7. Notably, we use this enhanced entropy module only in the heavyweight model, replacing the EM and CM in the framework shown in Figure 2. For the lightweight model, we use the standard entropy model from CompressAI.

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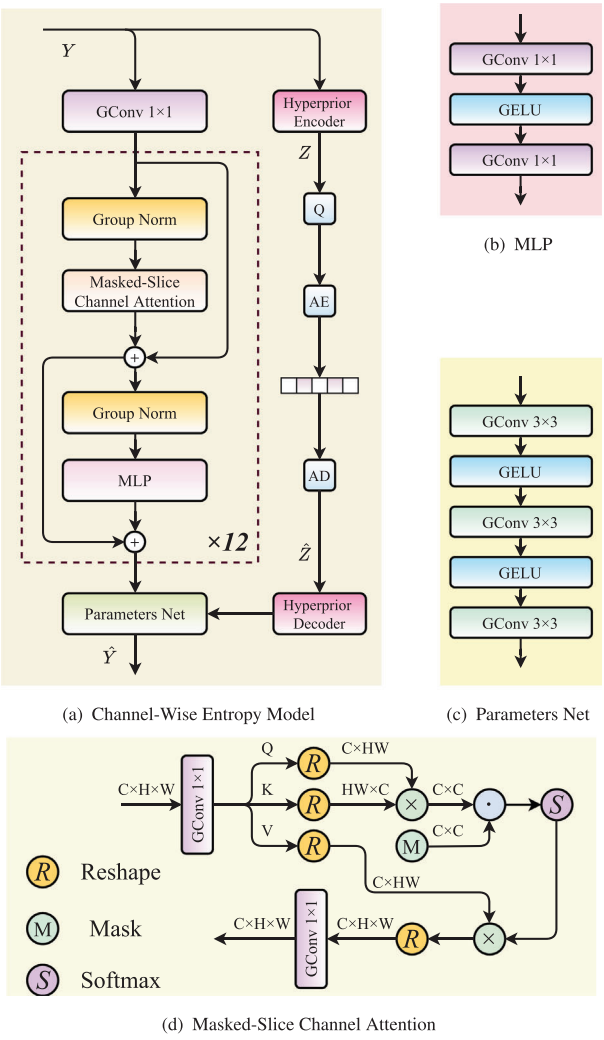


FIGURE 7 | The entropy model used for the heavyweight model.

In Figure 7a, the input feature $Y \in \mathbb{R}^{C\times H\times W}$, is sliced into N parts, each containing C/N channels. The sliced results are processed by the masked-slice channel attention module, where independent processing of each slice is achieved through group convolution and group normalization. Similar to the Vision Transformer [38], the core attention module is followed by an MLP module to enhance the network's representation capacity. Finally, after stacking 12 layers, the output is passed to the Parameters Net, which predicts the mean and variance for entropy encoding.

3.6 | Loss Function

The loss function of the network is defined by the rate-distortion optimization function, which balances visual quality and compression rate to achieve optimal performance. As introduced in Section 2, the loss function can be expressed as:

$$L = R(\hat{Y}) + \lambda \cdot d(X, \hat{X}), \tag{15}$$

where X is the original image, $\hat{X}$ is the reconstructed image, and $\hat{Y}$ is the quantized latent representation of the original image generated by the encoder. The rate $R(\hat{Y})$ is computed using the entropy of $\hat{Y}$, which is estimated by the hyperprior encoder and decoder. The d is distortion which can be measured by MSE or MS-SSIM. The hyperparameter λ is a hyperparameter that controls the degree of compression.

4 | Experiment

In this section, we first introduce the relevant experimental settings. Subsequently, we evaluate the performance of the proposed compression model and compare it with recent methods to demonstrate its competitive performance. Finally, we analyse the computational complexity of the proposed model, showing that it achieves comparable results with a modest reduction in computational costs.

4.1 | Settings

We constructed a high-resolution image dataset to train our proposed model. This dataset comprises 800 images from the DIV2K dataset [51], 585 images from the CLIC2020 dataset [52], 2650 images from the Flickr2K dataset [53], and 20,745 images from the Flickr2W dataset [54], totalling 24,780 high-resolution natural images. During training, images are randomly cropped into 256×256 patches each epoch. The test set consists of three datasets: the Kodak dataset [55], which contains 24 images with a resolution of 768×512; the Tecnick dataset [56] containing 40 images with a resolution of 1200×1200; and 26 representative high-resolution images selected from the CLIC2020 test set [52]. These datasets include various types of natural images, such as portraits, food, and landscapes, and are widely used to evaluate image compression performance.

Our network utilizes the Adam [57] optimizer with an initial learning rate of 1×10^{-4}. Instead of manually setting the learning rate decay, we rely on the adaptive learning rate function provided by PyTorch. This function evaluates the network performance after several rounds of training, reducing the learning rate if no improvement is observed. Based on our experimental results, we recommend setting this number of rounds between 5 and 15, with a learning rate reduction factor of 0.5. As mentioned in the previous section, the bitrates control will regulate the compression bitrates. We set six bitrate values: 0.0018, 0.0035, 0.0067, 0.0130, 0.0250, and 0.0483, consistent with CompressAI [58]. In this experiment, we use mean squared error (MSE) as the distortion metric in the loss function.

4.2 | Compression Performance

We conducted a quantitative evaluation of compression performance on the Kodak dataset [55], Tecnick dataset [56], and CLIC2020 dataset [52]. Following mainstream image compression evaluation methods, we use rate-distortion curves to describe compression performance. The rate-distortion curve plots distortion (measured by mean squared error (MSE) and multiscale structure similarity (MS-SSIM)) on the vertical axis and bitrates (measured by bits per pixel (BPP)) on the horizontal axis. We compare our proposed model with traditional compression methods such as JPEG [1], WebP [3], BPG [4], AVIF [6], and

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VTM [5], as well as mainstream autoencoder-based learned image compression schemes, including Ballé2016 [8], Ballé2018 [40], Minnen-mean [41], Minnen [41], Cheng2020 [59], Chen [34], GAACNN [17], and LIC2023 [21]. Please note that the Minnen-mean method, which will appear in Figure 8, also comes from Minnen et al. [41], but it does not include the contextual model.

Figure 8 presents the rate-distortion optimization curve. Based on our investigation, Chen et al. [34], Tang et al. [17], Cheng et al. [59], and Liu et al. [21] are recent autoencoder-based learned image compression schemes.

Under the MSE metric, our lightweight model shows a performance gap compared to recent mainstream methods, particularly at high bitrates. However, our heavyweight model performs significantly better, approaching the performance of the state-of-the-art methods [21].

Under the MS-SSIM metric, both our lightweight and heavyweight models demonstrate competitive performance, although there is still a gap compared to the state-of-the-art methods. This performance can be attributed to the introduction of Vision Mamba, which enables long-range modelling of image sequences and captures distant pixel relationships. By comprehensively considering both global and local feature extraction, our models achieve better results in structural similarity, making them highly competitive for high-quality image compression.

To further evaluate the compression performance, we provide a detailed comparison of BD-rate for these methods in Table A1. The table offers an intuitive assessment of the rate-distortion trade-offs, highlighting the efficiency of our models across different bitrates and metrics. In Table A1, our lightweight model achieves competitive performance compared to existing methods, while the heavyweight model attains near state-of-the-art performance.

In recent years, overfitted methods [27–29] have gradually gained popularity, and the latest research [29] shows that they can achieve performance superior to Ballé2016[8] and approaching Ballé2018 [40]. Based on our investigation, as of now, they have not yet reached the performance level of autoencoder-based methods. Overfitted methods, by leveraging their continuity, enable direct optimization of the rate-distortion trade-off. Besides, their parameter-efficient architectures have a higher theoretical performance ceiling compared to discrete coding frameworks. Therefore, they are considered more promising and potentially transformative approaches. Given the architectural incompatibility between continuous neural representations and VAE-based frameworks, a direct quantitative comparison falls beyond the scope of this study.

Figure 9 presents the entropy model capabilities of various methods through heatmaps. Models with stronger compression performance also exhibit better entropy modelling capabilities, as they accurately capture key features and precisely predict the mean and variance at each pixel location. By incorporating Vision Mamba to handle long-range dependencies, our lightweight model effectively preserves edge structures and detailed information in the mean and variance. The heavyweight model equipped with advanced entropy model, outperforms other methods. This is reflected in the heatmap, where the heavyweight model captures most of the image information, with details such as facial features being clearly visible in the mean predictions.

TABLE 1 | Comparison of algorithm complexity.

Method	FLOPs	Parameters
Ballé2016 [8]	12058.62M	2.89M
Ballé2018 [40]	12335.71M	4.97M
Minnen-mean [41]	12796.04M	6.92M
Minnen [41]	27038.65M	12.05M
Cheng2020 [59]	29968.33M	12.26M
Chen [34]	207249.41M	47.72M
GAACNN [17]	614748.50M	94.94M
LIC2023 [21]	116775.85M	75.90M
Our Model (light)	78402.87M	34.62M
Our Model (heavy)	119092.55M	105.37M

In Appendix, we examine the compression details using images from the Kodak and CLIC2020 datasets as examples. We compare our model with mainstream methods. The enlarged detail comparisons demonstrate that our method achieves better results in edge regions and significantly reduces compression artifacts (Figures A1, A2, and A3).

4.3 | Algorithm Complexity

In the study of learned image compression models, assessing algorithm complexity is crucial. Algorithm complexity directly impacts the training and inference speed of the model, as well as its deployability on different hardware platforms. Therefore, in this section, we will focus on introducing the algorithm complexity of our proposed model and compare it with other relevant methods. We present the FLOPs and parameters of various methods in Table 1, with the test tensor size set to 1×3 × 256×256. All the experiments in this section were conducted on an NVIDIA Tesla V100.

We excel in algorithm complexity according to Table 1. Although Tang et al. [17] have achieved the better rate-distortion performance in recent years, its floating point operations (FLOPs) are significantly higher than those of other models, nearly eight times that of our proposed model. Additionally, the high parameter count of their model makes it difficult to deploy. Similarly, Chen et al. [34] has FLOPs nearly three times higher than our model.

Table 1 lists the parameters of various methods. Our lightweight model has much lower parameters, making it more deployable. Our heavyweight model, while achieving the best compression performance at the cost of increased parameters, still maintains an acceptable level of FLOPs despite its larger parameters count.

In image compression, encoding and decoding time are critical metrics for evaluating compression performance. Table 2 compares the encoding and decoding times of several mainstream learned image compression methods. Due to the additional

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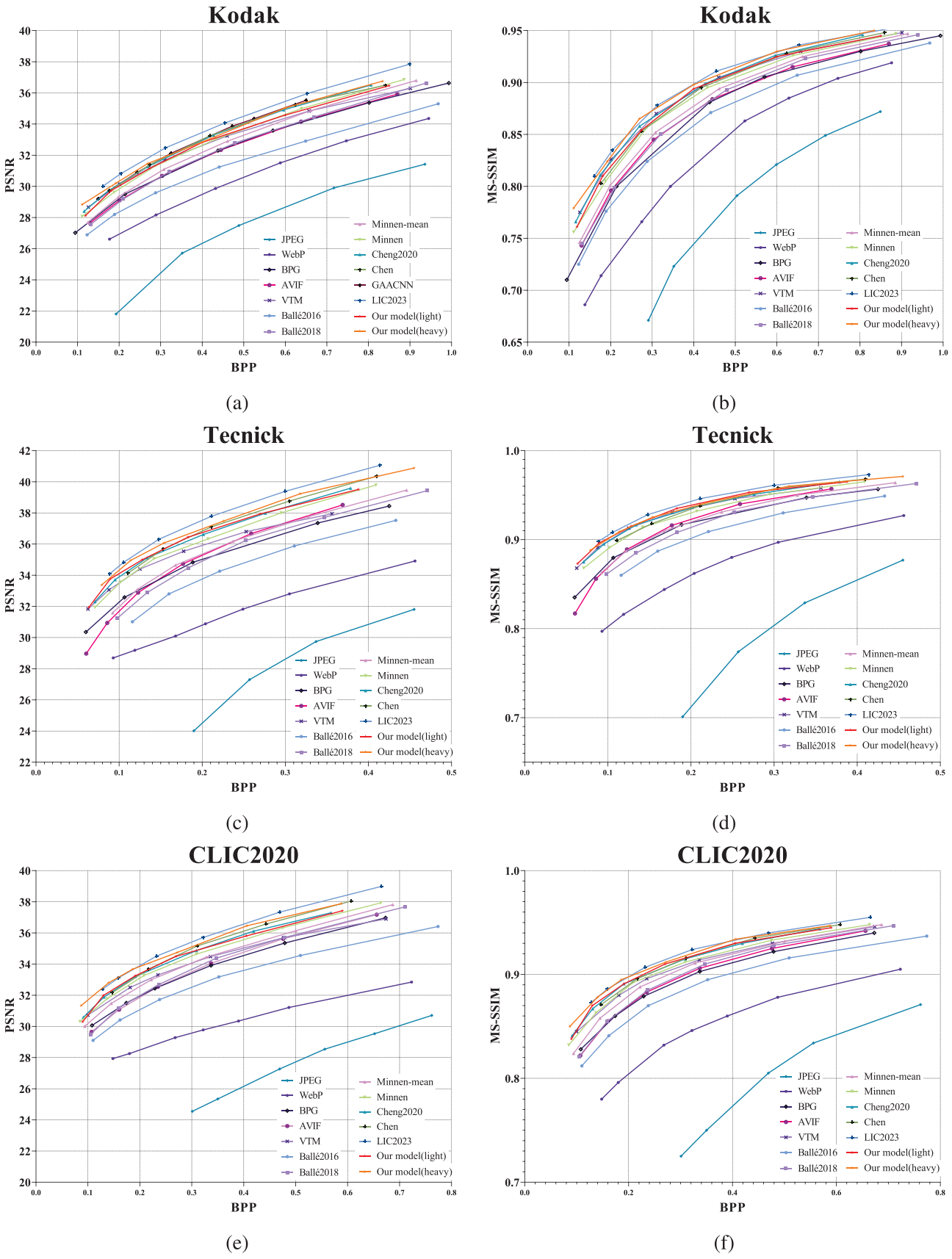


FIGURE 8 | Rate-distortion performance evaluation of various image compression methods on the Kodak, Tecnick, CLIC2020 dataset. (a) Distortion measured by MSE on Kodak. (b) Distortion measured by MS-SSIM on Kodak. (c) Distortion measured by MSE on Tecnick. (d) Distortion measured by MS-SSIM on Tecnick. (e) Distortion measured by MSE on CLIC2020. (f) Distortion measured by MS-SSIM on CLIC2020.

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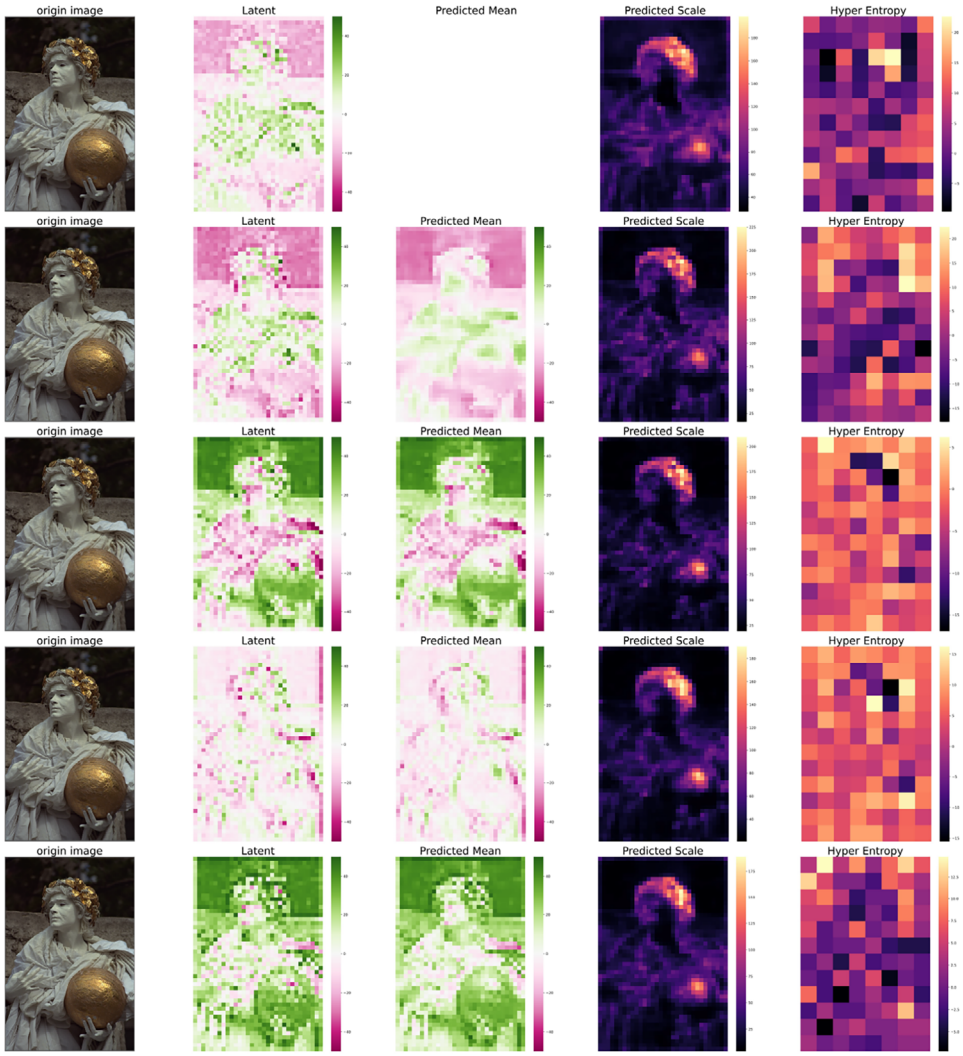


FIGURE 9 | Entropy model's modelling of data distribution presented in heatmap format. From the first row to the fourth row are the heatmaps of Ballé2018 [40], Minnen-mean [41], Minnen [41], our proposed lightweight model and heavyweight model respectively. Ballé2018 [40] does not have a predicted mean because it uses zero-mean Gaussian modelling.

TABLE 2 | Average running time of various methods.

Method	Encoding time (s)	Decoding time (s)
Ballé2016 [8]	0.06	0.23
Ballé2018 [40]	0.07	0.22
Minnen-mean [41]	0.08	0.26
Minnen [41]	3.50	8.59
Cheng2020 [59]	7.56	9.18
Chen [34]	2.31	353.27
GAACNN [17]	2.11	341.28
LIC2023 [21]	0.48	0.46
Our Model (light)	3.64	8.87
Our Model (heavy)	311.21	303.17

modules in our network framework compared to Minnen et al. [41], the encoding and decoding times are slightly longer. Chen et al. [34] and Tang et al. [17] perform well in encoding time but have very long decoding times. Tang et al. [17] mentioned in their paper that this is due to the 3D CNN-based autoregressive contexts module consuming a significant amount of computational resources. However, our lightweight model adopts a more efficient autoregressive contexts module from CompressAI [58], avoiding this issue.

5 | Conclusion

In this work, we propose an end-to-end image compression model based on dynamic convolution and Vision Mamba. Based on our investigation, our work is one of the early attempts to adapt Vision Mamba for image compression, demonstrating its potential in this field. We leverage Vision Mamba to address the challenge of long-range dependencies caused by insufficient global information. Additionally, we embed dynamic convolution into the network backbone, enriching feature extraction methods to achieve more accurate image data modelling. Building on the design of a lightweight standard model, we further develop a heavyweight model utilizing a more advanced entropy model. Extensive

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experiments were conducted on benchmark datasets to evaluate and compare compression quality and encoding/decoding time. The results demonstrate that our proposed image compression model can achieve performance comparable to recent models while maintaining low complexity. Therefore, we believe that the Mamba also holds great potential for exploration in the field of image compression.

Author Contributions

Lingchen Qiu: investigation, methodology, software, validation, visualization, writing – original draft. **Enjian Bai:** conceptualization, project administration, supervision, writing – review and editing. **Yun Wu:** supervision, writing –review and editing. **Yuwen Cao:** supervision, writing –review and editing. **Xue-qin Jiang:** project administration, resources, supervision, writing – review and editing.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the author upon reasonable request.

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Appendix A: Additional Experimental Results

This appendix provides supplementary experimental results, including visual comparisons of reconstructed images and a detailed BD-rate comparison table.

TABLE A1 | BD-rate comparison of the proposed method and existing approaches, using Ballé2016 [8] as the reference. Evaluations are conducted on the Kodak, Tecnick, and CLIC2020 datasets. The best-performing method is highlighted in red, while the second-best method is highlighted in blue.

	Kodak		Tecnick40		CLIC2020	
Method	**MSE**	**MS-SSIM**	**MSE**	**MS-SSIM**	**MSE**	**MS-SSIM**
JPEG [1]	145.33%	121.99%	246.72%	237.15%	390.02%	258.14%
WebP [3]	25.16%	33.04%	86.35%	63.97%	132.58%	81.82%
BPG [4]	−22.64%	−9.17%	−24.46%	−23.96%	−22.22%	−16.46%
AVIF [6]	−21.49%	−10.99%	−27.43%	−27.67%	−24.36%	−19.88%
VTM [5]	−34.10%	−28.24%	−39.62%	−44.73%	−33.12%	−32.39%
Ballé2016 [8]	0%	0%	0%	0%	0%	0%
Ballé2018 [40]	−22.04%	−12.63%	−21.81%	−17.47%	−26.30%	−22.25%
Minnen-mean [41]	−27.86%	−17.03%	−28.57%	−24.80%	−33.59%	−28.71%
Minnen [41]	−34.52%	−23.65%	−42.58%	−38.02%	−41.06%	−35.61%
Chen [34]	−39.21%	−26.17%	−48.06%	−40.65%	−46.61%	−39.48%
Cheng2020 [59]	−38.66%	−29.29%	−47.47%	−44.12%	−44.82%	−39.89%
LIC2023 [21]	−45.35%	−33.55%	−56.48%	−50.44%	−53.13%	−46.11%
Our model (light)	−35.40%	−27.48%	−50.17%	−47.77%	−44.91%	−41.75%
Our model (heavy)	−39.21%	−32.97%	−52.63%	−45.80%	−51.49%	−45.22%

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FIGURE A1 | Comparison of reconstructed images generated by the proposed method and existing approaches.

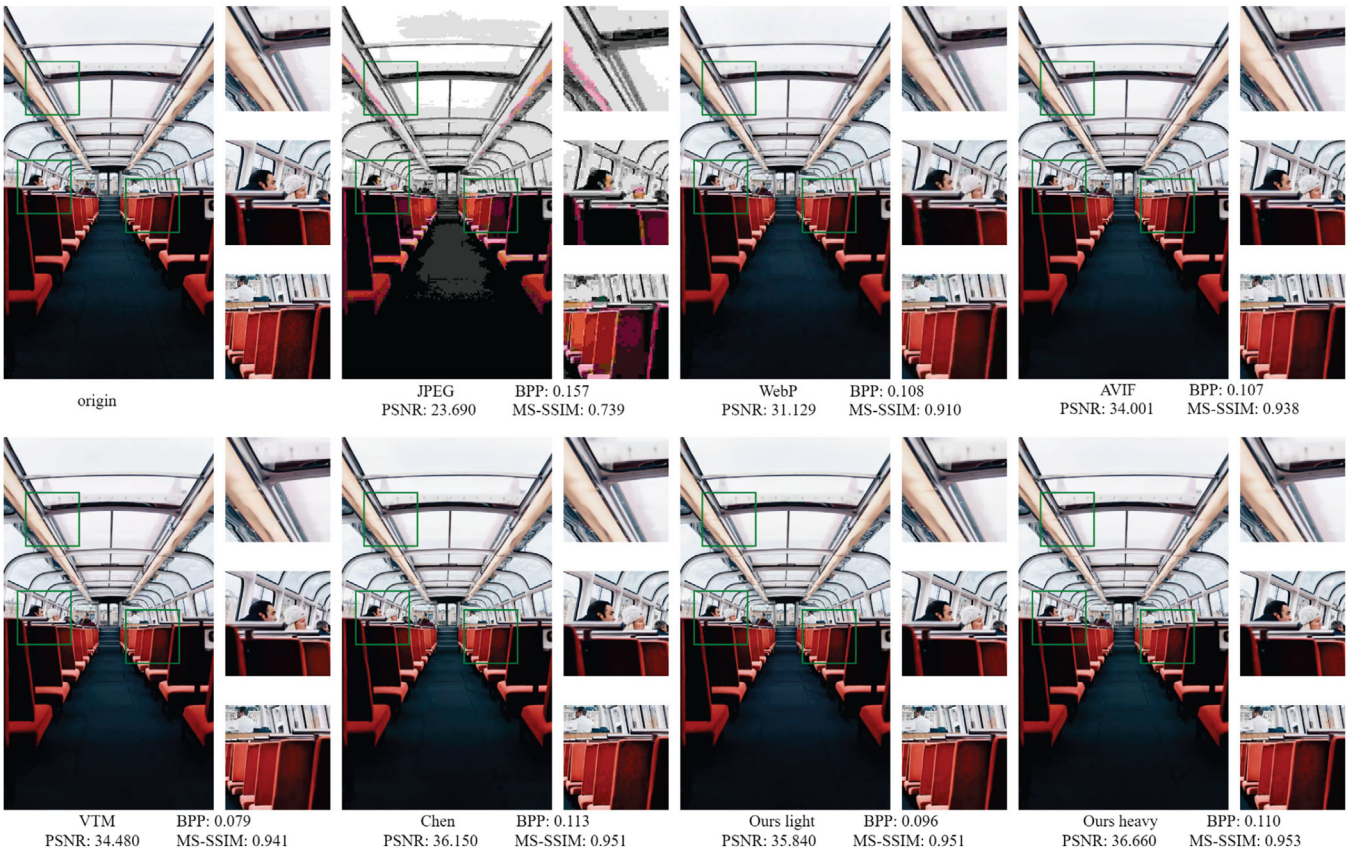


FIGURE A2 | Comparison of reconstructed images generated by the proposed method and existing approaches.

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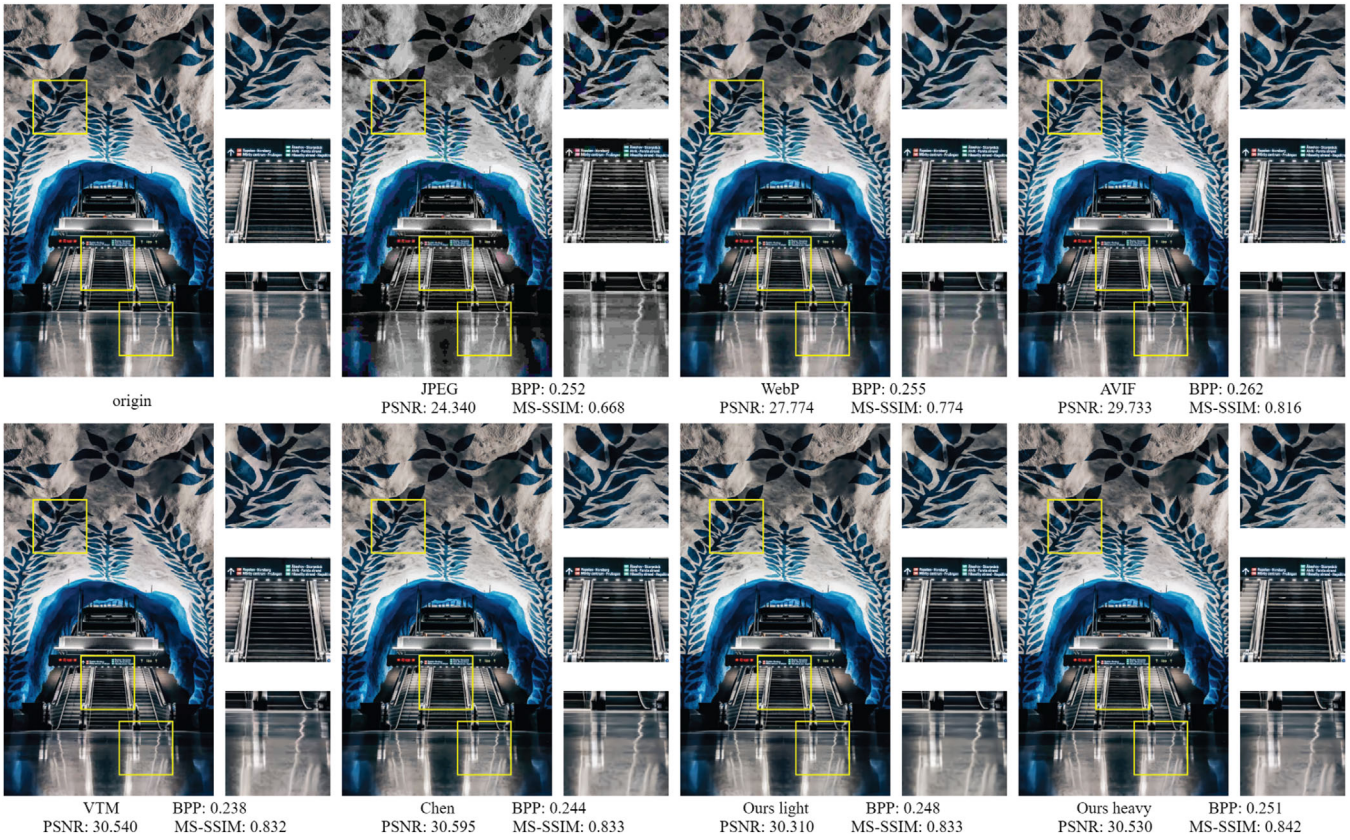


FIGURE A3 | Comparison of reconstructed images generated by the proposed method and existing approaches.

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